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PROFIT AND LOSS

3.1. What is Profit?

When a person does a business transaction and gets more than what he had invested, then he is said to have profit. The profit he gets will be equal to the additional money he gets other than his investment.

So profit can be understood as the extra money one gets other than what he had invested.

Example: A person bought an article for Rs. 100 and sold it for Rs. 120. Then he got Rs. 20 extra and so his profit is Rs. 20.

3.2. What is Loss?

When a person gets an amount less than what he had invested, then he is said to have a loss. The loss will be equal to the deficit he got than the investment.

Example: A person bought an article at Rs. 100 and sold it for Rs. 90. Then he got a deficit of Rs. 10 and so his loss is Rs. 10.

3.3. Cost Price (CP)

- The money that the trader puts in his business is called Cost Price. The price at which the articles are bought is called Cost Price.
- In other words, Cost Price is nothing but the investment in the business.

3.4 Selling Price (SP)

- The price at which the articles are sold is called the Selling Price. The money that the trader gets from the business is called Selling Price.
- In other words, Selling Price is nothing but the returns from a business.

3.5. Marked/Market/List Price (MP):

- The price that a trader marks or lists his articles to is called the Marked Price.
- This is the only price known to the customer.

3.6. Discount

The waiver of cost from the Marked Price that the trader allows a customer is called Discount.

**Note:**

1. Profit or loss percentage is to be applied always to the Cost Price only.
2. Discount percentage is to be applied always to the Marked Price only.

3.7. Relationship Among CP, SP and MP:

A trader adds his profit to the investment and sells it at that increased price.

Also he allows a discount on Marked Price and sells at the discounted price.

So, we can say that,

- $SP = CP + \text{Profit}$. (CP applied with profit is SP)
- $SP = MP - \text{Discount}$. (MP applied with discount is SP)

3.8. Understanding Profit and Loss:

So, by now we came to know that if CP is increased and sold it would result in profit and vice versa.

Also whatever increase is applied to CP, that increase itself is the profit.

For Rs. 10 profit, CP is to be increased by RS. 10 and the increased price becomes SP.

For 10% profit, CP is to be increased by 10% and it is the SP.

(From previous chapter we know that any value increased by 10% becomes 1.1 times.)

So, for 10% profit, CP increased by 10% $\Rightarrow 1.1CP = SP$.

- $SP = 1.1CP \Rightarrow \frac{SP}{CP} = 1.1 \Rightarrow 10\% \text{ profit}$
- $SP = 1.07CP \Rightarrow \frac{SP}{CP} = 1.07 \Rightarrow 7\% \text{ profit}$
- $SP = 1.545CP \Rightarrow \frac{SP}{CP} = 1.545 \Rightarrow 54.5\% \text{ profit and so on.}$

Similarly,

- $SP = 0.9CP \Rightarrow \frac{SP}{CP} = 0.9 \Rightarrow 10\% \text{ loss (Since 10\% decrease)}$
- $SP = 0.7CP \Rightarrow \frac{SP}{CP} = 0.76 \Rightarrow 24\% \text{ loss and so on.}$

So, to calculate profit % or loss %, it is enough for us to find the ratio of SP to CP.

Note:

1. If $SP/CP > 1$, it indicates profit.
2. If $SP/CP < 1$, it indicates loss.

3.9. Multiple Profits or Losses

A trader may sometimes have multiple profits or losses simultaneously. This is equivalent to having multiple changes and so all individual changes are to be multiplied to get the overall effect.



Examples: A trader uses a 800gm weight instead of 1 kg. Find his profit %.

Solution: (He is buying 800 gm but selling 1000 gm.)

So, CP is for 800 gm and SP is for 1000 gm.)

$$\frac{SP}{CP} = \frac{1000}{800} = 1.25 \Rightarrow 25\% \text{ profit.}$$

Examples: A trader uses 1 kg weight for 800 gm and increases the price by 20%. Find his profit/loss %.

Solution: 1 kg weight for 800 gm $\Rightarrow$ loss (decrease) $\Rightarrow 800/1000 = 0.8$

20% increase in price $\Rightarrow$ profit (increase) $\Rightarrow 1.2$

So, net effect = $(0.8) \times (1.2) = 0.96 \Rightarrow 4\% \text{ loss.}$

Examples: A milk vendor mixes water to milk such that he gains 25%. Find the percentage of water in the mixture.

Solution: To gain 25%, the volume has to be increased by 25%.

So, for 1 lt of milk, 0.25 lt of water is added $\Rightarrow$ total volume = 1.25 lt

$$\% \text{ of water} = \frac{0.25}{1.25} \times 100 = 20\% .$$

Examples: A trader bought an item for Rs. 200. If he wants a profit of 22%, at what price must he sell it?

Solution: CP=200, Profit = 22%.

So, $SP = 1.22CP = 1.22 \times 200 = 244/-.$

Examples: A person buys an item at Rs. 120 and sells to another at a profit of 25%. If the second person sells the item to another at Rs. 180, what is the profit % of the second person?

Solution: SP of 1st person = CP of 2nd person = $1.25 \times 120 = 150.$

SP of 2nd person = 180.

$$\text{Profit \%} = \frac{SP}{CP} = \frac{180}{150} = 1.2 \Rightarrow 20\% .$$

